

Family-wise error rate for multiple test points

Conformal prediction is typically presented as a method for constructing a prediction interval for a single test point X_{n+1} , given a training set of size n . However, it is often of interest to provide prediction intervals for multiple test points in a batch. This is a question of multiplicity: do we need to adjust our construction of the prediction sets to account for the fact that we are performing inference for multiple test points?

Throughout this section, we will use our previous notation for the split conformal prediction setting: we write $\mathcal{D}_n = ((X_i, Y_i))_{i \in [n]}$ to denote the calibration set, while $s : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ is the pretrained score function. Given a set of test data points $X'_1, \dots, X'_m$, we might be interested in

$$\mathbb{P}(Y_1 \in \mathcal{C}(X_1)) \geq 1 - \alpha$$

obtaining a guarantee of the form

$$\mathbb{P}(Y'_1 \in \mathcal{C}(X'_1), \dots, Y'_m \in \mathcal{C}(X'_m)) \geq 1 - \alpha_{\text{FWER}}, \quad (10.7)$$

meaning *all* of the m prediction sets cover (with probability at least $1 - \alpha_{\text{FWER}}$). Here α_{FWER} is a bound on the family-wise error rate: the probability of *at least one* miscoverage among our m prediction sets.

Let us now consider how the split conformal prediction method would perform relative to this goal. Since the prediction set is given by $\mathcal{C}(x) = \{y \in \mathcal{Y} : s(x, y) \leq \hat{q}\}$, where $\hat{q} = \text{Quantile}(S_1, \dots, S_n; (1 - \alpha)(1 + 1/n))$, we can rephrase the goal (10.7) above as

$$\mathbb{P}(S'_1 \leq \hat{q}, \dots, S'_m \leq \hat{q}) \geq 1 - \alpha_{\text{FWER}},$$

where the test point scores are given by $S'_i = s(X'_i, Y'_i)$. The key challenge is that the events $\{S'_i \leq \hat{q}\}_{i \in [m]}$ are not independent, since the conformal quantile $\hat{q}$ (which is random) provides a threshold that is shared across all m events.

In this next result, we consider both the finite-sample setting, where the number of calibration points n is fixed, and an asymptotic regime, where the number of calibration points is $n \rightarrow \infty$ (while the number of test points m , and the pretrained score function s , remain fixed). To accommodate both of these settings, we will write $\hat{q}_n$ (rather than $\hat{q}$) to emphasize that n may be increasing.

Proposition 10.2: The FWER of split conformal prediction (asymptotic)

Let $(X'_1, Y'_1), \dots, (X'_m, Y'_m), (X_1, Y_1), (X_2, Y_2), \dots$ be sampled i.i.d. from some distribution P . Let $s : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ be a pretrained score function, and for each calibration set size $n \geq 1$, define the split conformal prediction set as

$$\mathcal{C}_{\hat{q}_n}(x) = \{y : s(x, y) \leq \hat{q}_n\}$$

where

$$\hat{q}_n = \text{Quantile}(S_1, \dots, S_n; (1 - \alpha)(1 + 1/n))$$

for scores $S_i = s(X_i, Y_i)$, $i \in [n]$. Define $q^* = \text{Quantile}(F, 1 - \alpha)$, where F is the CDF of $s(X, Y)$ when $(X, Y) \sim P$.

Then for any fixed $n, m \geq 1$,

$$\mathbb{P}(Y'_1 \in \mathcal{C}_{\hat{q}_n}(X'_1), \dots, Y'_m \in \mathcal{C}_{\hat{q}_n}(X'_m)) \geq (1 - \alpha)^m.$$

$$\geq 1 - \alpha_{\text{FWER}} = 0.95$$

Moreover, if $F(q)$ is continuous and strictly increasing at $q = q^*$, then for any fixed $m \geq 1$,

$$\lim_{n \rightarrow \infty} \mathbb{P}(Y'_1 \in \mathcal{C}_{\hat{q}_n}(X'_1), \dots, Y'_m \in \mathcal{C}_{\hat{q}_n}(X'_m)) = (1 - \alpha)^m.$$

Proof of Proposition 10.2

By construction of the split conformal prediction set, $Y'_i \in \mathcal{C}_{\hat{q}_n}(X'_i)$ holds if and only if $s(X'_i, Y'_i) \leq \hat{q}_n$. Since the test point scores $S'_i = s(X'_i, Y'_i)$ are i.i.d. draws from the CDF F , and $\hat{q}_n$ depends only on the calibration data (and is therefore independent of the test scores), we have

$$\mathbb{P}(S'_1 \leq \hat{q}_n, \dots, S'_m \leq \hat{q}_n) = \mathbb{E}[\mathbb{P}(S'_1 \leq \hat{q}_n, \dots, S'_m \leq \hat{q}_n \mid \hat{q}_n)] = \mathbb{E}[F(\hat{q}_n)^m].$$

The function $t \mapsto t^m$ is convex on $t \geq 0$, so $\mathbb{E}[F(\hat{q}_n)^m] \geq \mathbb{E}[F(\hat{q}_n)]^m$ by Jensen's inequality. Finally, by definition of F , we can write $F(\hat{q}_n) = \mathbb{P}(s(X'_1, Y'_1) \leq \hat{q}_n \mid \hat{q}_n) =$

$\mathbb{P}(Y'_1 \in \mathcal{C}_{\hat{q}_n}(X'_1) \mid \hat{q}_n)$, where $(X'_1, Y'_1) \sim P$ is a test point. Thus, by the tower law,

$$\mathbb{E}[F(\hat{q}_n)] = \mathbb{E}[\mathbb{P}(Y'_1 \in \mathcal{C}_{\hat{q}_n}(X'_1) \mid \hat{q}_n)] = \mathbb{P}(Y'_1 \in \mathcal{C}_{\hat{q}_n}(X'_1)) \geq 1 - \alpha,$$

where the last step holds by the distribution-free marginal coverage guarantee for the split conformal prediction method. This completes the proof for the finite-sample case.

Next, assume $F(q)$ is continuous at $q = q^*$, and that q^* is the unique $(1 - \alpha)$ -quantile, then for any fixed $m \geq 1$. By Theorem 5.6, it holds almost surely that $\hat{q}_n \rightarrow q^*$. Since F is assumed to be continuous at q^* , we then have $F(\hat{q}_n) \rightarrow F(q^*) = 1 - \alpha$, almost surely. By the dominated convergence theorem, then, it holds that $\mathbb{E}[F(\hat{q}_n)^m] \rightarrow (1 - \alpha)^m$, which proves the desired result for the asymptotic setting.

This proposition reveals a fundamental fact about FWER control with conformal prediction: it is not possible to do much better than a union bound. That is, the aim of FWER control, as stated in (10.7), will only hold (asymptotically) if we choose

$$1 - \alpha_{\text{FWER}} = (1 - \alpha)^m \iff \alpha = 1 - (1 - \alpha_{\text{FWER}})^{1/m} \approx \alpha_{\text{FWER}}/m,$$

i.e., this is nearly the Bonferroni correction. This is because, as the calibration set size n approaches infinity, the events $\{S'_i \leq \hat{q}_n\}_{i \in [m]}$ become approximately independent.

$$\approx \frac{0.05}{m}$$

Test $m = 1000$

$$\approx \frac{0.05}{1000}$$

Outlier detection with false discovery rate control

In this section, we turn to a different setting where multiplicity arises: the problem of outlier detection. As before, assume we have i.i.d. calibration points $(X_1, Y_1), \dots, (X_n, Y_n)$ from a distribution P , and an independent set of test points $(X'_1, Y'_1), \dots, (X'_m, Y'_m)$. In contrast to the prediction problem, our goal now is to detect outliers (i.e., test points (X'_i, Y'_i) that were *not* drawn from P), rather than to construct prediction intervals. In particular, $Y'_1, \dots, Y'_m$ are observed, which was not previously the case.

To formalize the question, for each test point $i \in [m]$ define the null hypothesis

$$H_{0,i} : (X'_i, Y'_i) \sim P.$$

Rejecting $H_{0,i}$ means that we claim to have detected that test point (X'_i, Y'_i) is an outlier. For the outlier detection problem, when we are simultaneously testing m many data points, we generally wish to detect outliers without incurring too many false positives. In particular, consider a procedure that sees both the calibration points and the test points and then returns a subset $\mathcal{R} \subseteq \{1, \dots, m\}$ of the test points that are believed to be outliers—this is the set of *rejections* among the null hypotheses $H_{0,1}, \dots, H_{0,m}$ being tested.

$$\mathcal{D}_{\text{cal}}, \mathcal{D}_{\text{test}} \quad \mathcal{R} = \{i : H_{0,i} \text{ is rejected}\}$$

When the number of test points m is large, a natural notion of error in this setting is the false discovery rate (FDR), defined as

$$\text{FDR} = \mathbb{E} \left[\frac{\sum_{i=1}^m \mathbb{1}\{i \in \mathcal{R}, H_{0,i} \text{ is true}\}}{\max\{1, |\mathcal{R}|\}} \right], \leq \alpha$$

the expected proportion of false rejections.

We now consider a conformal p-value approach towards testing the m null hypotheses with FDR control. Assume that we have a pretrained score function $s : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$. We will test each null hypothesis $H_{0,i}$ by comparing the score $S'_i = s(X'_i, Y'_i)$ against the calibration set scores $S_1, \dots, S_n$. Concretely, we consider the conformal p-value for test point i (recall Section 3.5.1):

$$p_i = \frac{1 + \sum_{j=1}^n \mathbb{1}\{S'_i \leq S_j\}}{n + 1}. \quad (10.8)$$

$$p_1, \dots, p_m \Rightarrow \mathcal{R} \text{ s.t. } \text{FDR} \leq \alpha$$

Notice that the p-values $(p_1, \dots, p_m)$ are dependent, because they are all calculated using the same calibration set. This is the main challenge we address in this section—we will characterize the dependence structure and explain how this enables the use of existing FDR control algorithms together with these conformal p-values.

FDR control with the Benjamini–Hochberg procedure

Controlling the FDR is well understood if we have independent p-values. For example, the widely-used Benjamini–Hochberg algorithm is defined as

$$\mathcal{R} = \left\{ i : p_i \leq \frac{\alpha_{\text{FDR}} \hat{k}}{m} \right\} \text{ where } \hat{k} = \max \left\{ k : \sum_{i=1}^m \mathbb{1} \left\{ p_i \leq \frac{\alpha_{\text{FDR}} k}{m} \right\} \geq k \right\}, \quad (10.9)$$

where $\alpha_{\text{FDR}} \in [0, 1]$ is the target FDR level. When applied to independent p-values, this algorithm has the property that $\text{FDR} \leq \alpha_{\text{FDR}}$.

Equivalent Formulations of the Benjamini–Hochberg Procedure

Let $p_1, \dots, p_m \in [0, 1]$ be the p-values, and write $p_{(1)} \leq \dots \leq p_{(m)}$ for their ordered values. For a target level $\alpha_{\text{FDR}} \in (0, 1)$, the Benjamini–Hochberg (BH) procedure selects the largest admissible rank,

$$\hat{k} = \max \left(\{0\} \cup \left\{ k \in \{1, \dots, m\} : p_{(k)} \leq \frac{\alpha_{\text{FDR}} k}{m} \right\} \right), \quad (1)$$

and rejects the hypotheses corresponding to the $\hat{k}$ smallest p-values. If $\hat{k} = 0$, no hypotheses are rejected.

To see why this agrees with the counting formulation, observe that, for every $k \in \{1, \dots, m\}$ and threshold t ,

$$p_{(k)} \leq t \iff \sum_{i=1}^m \mathbb{1}\{p_i \leq t\} \geq k.$$

Indeed, the k th smallest p-value is at most t precisely when at least k of the p-values are at most t . Taking $t = \alpha_{\text{FDR}} k / m$ gives

$$\hat{k} = \max \left(\{0\} \cup \left\{ k \in \{1, \dots, m\} : \sum_{i=1}^m \mathbb{1} \left\{ p_i \leq \frac{\alpha_{\text{FDR}} k}{m} \right\} \geq k \right\} \right). \quad (2)$$

$\hat{k} = 3$ reject $p_{(1)}, p_{(2)}, p_{(3)}$

$R = \{$

The rejection set can therefore be written as

$$\mathcal{R} = \left\{ i \in \{1, \dots, m\} : p_i \leq \frac{\alpha_{\text{FDR}} \hat{k}}{m} \right\}. \quad (3)$$

For $\hat{k} \geq 1$, this threshold includes at least the $\hat{k}$ smallest p -values. If it included $r > \hat{k}$ p -values, then

$$p_{(r)} \leq \frac{\alpha_{\text{FDR}} \hat{k}}{m} \leq \frac{\alpha_{\text{FDR}} r}{m},$$

contradicting the maximality of $\hat{k}$. Hence $|\mathcal{R}| = \hat{k}$, and the two formulations produce exactly the same rejections. When $\hat{k} = 0$, the failure of the condition at $k = 1$ implies $p_{(1)} > \alpha_{\text{FDR}}/m > 0$, so the rejection set in (3) is empty.

This equivalence also holds when p -values are tied and requires no independence assumption; distributional assumptions are needed to establish FDR control, rather than the equivalence of the algorithms.

This leads to the main idea of this section: for the outlier detection problem, even though the conformal p -values p_i defined above are not independent, we can still control the FDR using the Benjamini–Hochberg algorithm. This is because the conformal p -values have a particular form of dependence:

$$p_1, \dots, p_m \xrightarrow{\text{BH}} \mathcal{R} = \left\{ i : p_i \leq \frac{\alpha_{\text{FDR}} \hat{k}}{m} \right\}$$

↑

$$\text{FDR} \leq \alpha_{\text{FDR}}$$

$$W = (P_1, \dots, P_m) \quad \mathbb{P}(W \in A | P_i = t)$$

Definition 10.3: Positive regression dependence on a subset (PRDS)

A random vector $W \in \mathbb{R}^m$ satisfies the PRDS property with respect to a set $I_0 \subseteq \{1, \dots, m\}$ if for any $i \in I_0$ and any nondecreasing set A , $\mathbb{P}(W \in A | W_i = w_i)$ is nondecreasing in w_i .

Here, a nondecreasing set $A \subseteq \mathbb{R}^m$ is a set where if $(a_1, \dots, a_m) \in A$, then any vector $(b_1, \dots, b_m)$, with $b_i \geq a_i$ for all $i \in [m]$, must also lie in A . Importantly, the Benjamini–Hochberg algorithm is known to control the FDR under this type of dependence—i.e., for PRDS p-values, the Benjamini–Hochberg procedure (10.9) guarantees $\text{FDR} \leq \alpha_{\text{FDR}}$.

$$D_{\text{cal}} \quad D_{\text{test}} \quad b > a \quad a \in A \Rightarrow b \in A \\ b_i \geq a_i \quad \forall i$$

We now state the main result: the PRDS property of conformal p-values constructed with a shared calibration set.

Theorem 10.4: Conformal p-values are PRDS

Suppose $(X_1, Y_1), \dots, (X_n, Y_n), (X'_1, Y'_1), \dots, (X'_m, Y'_m)$ are independent. Assume also that $(X_i, Y_i) \sim P$ for all $i \in [n]$, and that $(X'_i, Y'_i) \sim P$ for all $i \in I_0$. Let s be a pretrained score function, and assume that all calibration and test scores are distinct almost surely. Then, the conformal p-values (10.8) are PRDS on the subset I_0 .

$$\{x: x < 3\}$$

$$\{x: x \geq 3\}$$

In this setting, the p-values are no longer independent, but the result of Theorem 10.4 establishes that their dependence must be positive. Indeed, Figure 10.2 gives an intuitive illustration of why this should be the case: we see positive dependence among the conformal p-values $p_1, \dots, p_m$ because they are each comparing against the same reference set of scores (i.e., the scores of the calibration set). For example, if the calibration set happens to contain an unusually high fraction of data points (X_i, Y_i) with a low score $S_i = s(X_i, Y_i)$, then all p-values $p_1, \dots, p_m$ will be more likely to be small.

The practical implication of this theorem is that the Benjamini–Hochberg algorithm with the conformal p-values controls the FDR. We record this fact in the following corollary.

Corollary 10.5: FDR control with conformal p-values

In the setting above, let $\mathcal{R}$ be the set of rejections of the Benjamini–Hochberg algorithm, when run at target FDR level α_{FDR} based on the p-values $p_1, \dots, p_m$. Then, the FDR is controlled:

$$\mathbb{E} \left[\frac{|\{i \in \mathcal{R}, H_{0,i} \text{ is true}\}|}{\max\{1, |\mathcal{R}|\}} \right] \leq \alpha_{\text{FDR}}.$$

$p_1, \dots, p_m$ satisfy "PRDS"

Lastly, we remark that the PRDS result of Theorem 10.4 is not unique to the outlier detection setting and holds for split conformal p-values in general, but it is particularly important in the setting of outlier detection since it enables FDR control.

Proof of Theorem 10.4

Fix any test point in I_0 —without loss of generality we can assume it is the first one, (X'_1, Y'_1) . First, for each $i \in \{2, \dots, m\}$, we define a count

$$C_i = \sum_{j=1}^n \mathbb{1}\{s(X'_i, Y'_i) \leq s(X_j, Y_j)\} + \mathbb{1}\{s(X'_i, Y'_i) \leq s(X'_1, Y'_1)\}.$$

For intuition, we can observe that, for each $i \in \{2, \dots, m\}$, the quantity $\frac{1+C_i}{n+2}$ is the conformal p-value we would obtain for test point (X'_i, Y'_i) if we were comparing to an augmented calibration set $((X_1, Y_1), \dots, (X_n, Y_n), (X'_1, Y'_1))$ rather than only comparing to $\mathcal{D}_n = ((X_1, Y_1), \dots, (X_n, Y_n))$.

Step 1: verifying independence of p_1 and the C_i 's. First, we will prove that $p_1 \perp\!\!\!\perp (C_2, \dots, C_m)$. This claim holds by exchangeability of the augmented calibration set $(X_1, Y_1), \dots, (X_n, Y_n), (X'_1, Y'_1)$ —the counts C_i depend symmetrically on this augmented dataset, while p_1 expresses the rank of (X'_1, Y'_1) within this dataset and is therefore uniformly distributed over $\{\frac{1}{n+1}, \dots, \frac{n}{n+1}, 1\}$, even after conditioning on $C_2, \dots, C_m$, since we have assumed that the scores are distinct almost surely. (We

of augmented scores $\geq s'_i$
 $s_1 \dots s_n, s'_1$

$h(C_1) \dots h(C_m)$

$p_i = h(C_i)$
 intermediate

$(X_1, Y_1) \dots (X_n, Y_n), (X'_1, Y'_1)$
 cal test

$$p_i = \frac{1 + \sum_{j=1}^n \mathbb{1}(s(X_j, Y_j) \geq s(X'_i, Y'_i))}{1 + n}$$

$$P_1 \perp\!\!\!\perp C_2 \dots C_m$$

$$P_i = h(C_i)$$

can observe that this argument is nearly identical to that of Theorem 8.2, which proved independence of conformal p-values in the online testing regime.)

Step 2: recovering p_i 's from C_i 's. Next, we claim that for each $i \in \{2, \dots, m\}$,

$$C_i \leftarrow f\left(\frac{s(X'_i, Y'_i)}{s(X'_1, Y'_1)}\right) \iff C_i \geq (n+1)p_i \rightarrow (n+1)p_i$$

To see why, define $f(t) = \sum_{j=1}^n \mathbb{1}\{t \leq s(X_j, Y_j)\} + \mathbb{1}\{t \leq s(X'_1, Y'_1)\}$. Then by construction, $C_i = f(s(X'_i, Y'_i))$ for each $i \in \{2, \dots, m\}$, and $p_1 = \frac{f(s(X'_1, Y'_1))}{n+1}$. Since f is monotone, and is strictly decreasing over score values, this verifies the claim. With this calculation in place, we will now see that the original p-values p_i can be recovered as functions of p_1 and C_i : for each $i \in \{2, \dots, m\}$, by definition of p_i and of C_i , we have

$$p_i = \frac{1 + \sum_{j=1}^n \mathbb{1}\{s(X'_i, Y'_i) \leq s(X_j, Y_j)\}}{n+1} = \frac{1 + C_i - \mathbb{1}\{s(X'_i, Y'_i) \leq s(X'_1, Y'_1)\}}{n+1}$$

$$\begin{matrix} h(C_i) & h(C_m) & = \frac{1 + C_i - \mathbb{1}\{C_i \geq (n+1)p_1\}}{n+1} \end{matrix}$$

Step 3: the PRDS condition. Now let $A \subseteq \mathbb{R}^m$ be an nondecreasing set. We need to verify that

$$t \mapsto \mathbb{P}((p_1, \dots, p_m) \in A \mid p_1 = t)$$

is nondecreasing in t . From Step 2, this function can equivalently be written as

$$t \mapsto \mathbb{P}(V_t \in A \mid p_1 = t)$$

where we define the random vector V_t as

$$V_t = \left(t, \frac{1 + C_2 - \mathbb{1}\{C_2 \geq (n+1)t\}}{n+1}, \dots, \frac{1 + C_m - \mathbb{1}\{C_m \geq (n+1)t\}}{n+1} \right).$$

Since we have proved that $p_1 \perp\!\!\!\perp (C_2, \dots, C_m)$ in Step 1, while V_t is a function of $(C_2, \dots, C_m)$, we have $\mathbb{P}(V_t \in A \mid p_1 = t) = \mathbb{P}(V_t \in A)$. Therefore, we now only need to verify that

$$t \mapsto \mathbb{P}(V_t \in A)$$

is nondecreasing in t . Next, we can observe that $t \mapsto V_t$ is always a coordinatewise monotone nondecreasing function (i.e., this holds for any value of $C_2, \dots, C_m$). Since A is a nondecreasing set, then, it holds almost surely that

$$V_t \in A \implies V_{t'} \in A,$$

for any $t \leq t'$. Therefore, $t \mapsto \mathbb{P}(V_t \in A)$ is an nondecreasing function, as desired.

$p_1, \dots, p_m$ are PRDS

$\Rightarrow$ BH controls FDR

Why p_1 is independent of $(C_2, \dots, C_m)$

The counts record how the augmented scores compare with the other test scores, but do not identify which augmented score belongs to test point 1. This symmetry is the basis of the independence claim.

Setup and assumptions. Write $S_j = s(X_j, Y_j)$ for the calibration scores and $S'_i = s(X'_i, Y'_i)$ for the test scores. Treat the pretrained score function s as fixed, conditioning on independent training data if needed. As in the theorem, assume that all observations are independent, the calibration observations and test point 1 follow P (so $H_{0,1}$ is true), and all scores are distinct almost surely. The other test observations need not follow P .

The conformal p -values and the augmented counts are

$$p_i = \frac{1 + \sum_{j=1}^n \mathbf{1}\{S_j \geq S'_i\}}{n+1}, \quad (4)$$

$$C_i = \sum_{j=1}^n \mathbf{1}\{S_j \geq S'_i\} + \mathbf{1}\{S'_1 \geq S'_i\}, \quad i = 2, \dots, m. \quad (5)$$

Including S'_1 in *every* count makes the counts symmetric in the augmented sample $(S_1, \dots, S_n, S'_1)$.

$$p_1 \perp\!\!\!\perp C_2, C_3 \quad S'_1$$

The label remains unknown even when the values are known. Now fix the sorted sample at $(9, 5, 2)$, with the same test scores $S'_2 = 4$ and $S'_3 = 7$. The counts $(C_2, C_3) = (2, 1)$ remain unchanged under all possible assignments of the label S'_1 :

Value assigned to S'_1	Remaining calibration scores	p_1	(C_2, C_3)
9	$\{5, 2\}$	$1/3$	$(2, 1)$
5	$\{9, 2\}$	$2/3$	$(2, 1)$
2	$\{9, 5\}$	1	$(2, 1)$

Conditional on these sorted values and the other test scores, the three rows are equally likely. Knowing the counts leaves uncertainty about both the sorted values and the label assignment; knowing $\mathcal{G}$ resolves the first uncertainty but preserves the second. It is the remaining uncertainty about the label that makes p_1 uniform.

$$\{S_1, S_2, S'_1\} = \{9, 5, 2\} \quad \text{Augmented}$$

$$\frac{\sum_{j=1,2} \mathbf{1}(S_j \geq S'_1) + 1}{2+1} = \frac{1}{3}$$

$$(S_1, S_2, \dots, S_n, S'_1)$$

What remains random after the score values are known? Let $N = n + 1$, and sort the augmented scores in decreasing order: $n+1$

$$A_{[1]} > A_{[2]} > \dots > A_{[N]}.$$

Let $\mathcal{G}$ denote the information in this sorted list and the other test scores $(S'_2, \dots, S'_m)$. It does not specify which sorted entry is S'_1 . Because $S_1, \dots, S_n, S'_1$ are exchangeable, each assignment of their labels to the sorted values is equally likely. The other test scores are independent of this sample, so conditioning on them preserves this symmetry. Thus, conditional on $\mathcal{G}$, the label S'_1 is equally likely to occupy any of the N positions.

If $S'_1 = A_{[r]}$, exactly $r - 1$ calibration scores exceed it, and hence $p_1 = r/N$. Therefore, for each $r \in \{1, \dots, N\}$,

$$\mathbb{P}\left(p_1 = \frac{r}{N} \mid \mathcal{G}\right) = \frac{1}{N} \quad \text{almost surely.} \quad (6)$$

$$\begin{aligned} \mathcal{G} &= \{ \overset{S'_1 \rightarrow ?}{A_{[1]} > \dots > A_{[N]}} + (S'_2, \dots, S'_m) \} \\ &\Rightarrow p_1 ? \xrightarrow{\text{given } \mathcal{G}} p_1 \in \left\{ \frac{1}{n+1}, \frac{2}{n+1}, \dots, \frac{n}{n+1}, 1 \right\} \\ \mathcal{G} &\Rightarrow C_2, \dots, C_m ? \end{aligned}$$

Why conditioning on all the counts preserves uniformity. Every count can be computed from $\mathcal{G}$ alone, since

$$C_i = \sum_{\ell=1}^N \mathbf{1}\{A_{[\ell]} \geq S'_i\}. \quad \mathcal{G} \Rightarrow C_i$$

Relabeling can change p_1 but leaves all counts unchanged. Since the count vector is a function of $\mathcal{G}$, the tower property applied to (6) gives

$$\begin{aligned} \mathbb{P}\left(p_1 = \frac{r}{N} \mid C_2, \dots, C_m\right) &= \mathbb{E}\left[\mathbb{P}\left(p_1 = \frac{r}{N} \mid \mathcal{G}\right) \mid C_2, \dots, C_m\right] \\ &= \frac{1}{N} = \mathbb{P}\left(p_1 = \frac{r}{N}\right). \end{aligned} \quad (7)$$

For every possible value of p_1 , the conditional and marginal probabilities agree. Hence p_1 is independent of the whole count vector:

$$p_1 \perp\!\!\!\perp (C_2, \dots, C_m).$$

What the counts reveal about the sorted scores

Knowing $(C_2, \dots, C_m)$ generally does not determine the full sorted scores $(A_{[1]}, \dots, A_{[N]})$, even if the other test scores are also known. Each C_i records only how many augmented scores are at least S'_i ; it does not record their exact values.

Different sorted samples can give the same counts. Take $N = 3$, $S'_2 = 4$, and $S'_3 = 7$. The following two sorted samples produce identical counts:

Sorted augmented scores	C_2 : number ≥ 4	C_3 : number ≥ 7
$(9, 5, 2)$	2	1
$(10, 6, 1)$	2	1

Thus, the sorted scores together with the other test scores determine the counts, but the reverse generally fails:

$$\underbrace{(A_{[1]}, \dots, A_{[N]}, S'_2, \dots, S'_m)}_{\text{information } \mathcal{G}} \longrightarrow (C_2, \dots, C_m).$$

Why the proof conditions on more information. The proof first establishes uniformity given $\mathcal{G}$, which provides more information than the counts. Even with all these score values known, the position carrying the label S'_1 remains uniform by exchangeability. Conditioning only on the counts then averages over the richer configurations consistent with those counts. Every such configuration has conditional rank probability $1/N$, almost surely, so their conditional average remains $1/N$:

$$\mathbb{P}\left(p_1 = \frac{r}{N} \mid C_2, \dots, C_m\right) = \mathbb{E}\left[\frac{1}{N} \mid C_2, \dots, C_m\right] = \frac{1}{N}.$$

This is the averaging step in (7). Recovering the sorted scores from the counts is unnecessary.